NAME

mbroutetime − Program to output a list of the times when a survey passes the waypoints of a planned survey route.

VERSION

Version 5.0

SYNOPSIS

mbroutetime --route-file=routefile {-Rroutefile}
[--format=format_id {-Fformat_id}
--input=file {-Ifile}
--output=waypointtimefile {-Owaypointtimefile}
--range-threshold=rangethreshold {-Urangethreshold}
--verbose {-V}
--help {-H}]

DESCRIPTION

mbroutetime −

MB-SYSTEM AUTHORSHIP

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OPTIONS

−−format=format_id −Fformat_id

Sets the format for the input swath data using **MBIO** integer format identifiers. This program uses the **MBIO** library and will read any swath format supported by **MBIO**. A list of the swath data formats currently supported by **MBIO** and their identifier values is given in the **MBIO** manual page. Default: *format* = 0.

−−help −H

This "help" flag cause the program to print out a description of its operation and then exit immediately.

−−input=file −Ifile

Sets the input filename. If *format* > 0 (set with the **−F** option) then the swath data contained in *infile* is read and processed. If *format* < 0, then *infile* is assumed to be an ascii file containing a list of the input swath data files to be processed and their formats. The program will read the data in each one of these files. In the *infile* file, each data file should be followed by a data format identifier, e.g.:

datafile1 11
datafile2 24

This program uses the **MBIO** library and will read any swath format supported by **MBIO**. A list of the swath data formats currently supported by **MBIO** and their identifier values is given in the **MBIO** manual page. Default: *infile* = "datalist.mb-1".

−−output=waypointtimefile −Owaypointtimefile
Sets the name of the output file listing, for each route waypoint reached, the longitude, latitude, and time (in decimal epoch seconds) at which the survey passed that waypoint. Default: *waypointtimefile* is the input file or datalist name (set with the **−I** option) with the suffix "_wpttime_d.txt" appended.

−−route−file=routefile −Rroutefile
Sets the name of the input **MBgrdviz** route file containing the planned survey waypoints against which the survey data are checked. This option is required.

−−range−threshold=rangethreshold −Urangethreshold
Sets the distance in meters that the survey navigation must come within a route waypoint for that waypoint to be considered reached. Default: *rangethreshold* = 25.0.

−−verbose −V
The **−V** option causes **mbroutetime** to print out status messages.

EXAMPLES

Suppose you are going to run the MBARI Mapping AUV on the Coaxial Segment of the Juan de Fuca Ridge. You have created a route file called Coaxial2009_1v3.rte using **mbgrdviz**. In order to create an MBARI AUV mission script for mission beginning with a spiral descent to an altitude of 50 m, followed by a survey run at a 75 m altitude, the following will suffice:

mbroutetime −I Coaxial2009_1v3.rte -A75/30/75 −B1 -G0 −MSL −N50 −R220/83/0.3 -O Coaxial2009_1v3.cfg -S1.5 −L30 −W100 −V

The resulting mission script has the following header:

```
# This MBARI Mapping AUV mission file has been generated
# by the MB-System program mbroutetime run by
# user <caress> on cpu <shepard> at <Thu Jul 30 11:36:47 PDT 2009>
#
# Mission Summary:
#    Route File:            Coaxial2009_1v3.rte
#    Mission File:          Coaxial2009_1v3.cfg
#    Distance:              79501.503455 (m)
#    Estimated Time:        57203 (s)  15.890 (hr)
#    Abort Time:            60067 (s)
#    Max battery life:      64800 (s)
#    Safety margin:         1800 (s)
#    Ascend time:           2932 (s)
#    Way Points:            44
#    Route Points:          550
#    Survey behavior:       WaypointDepth
#    Descent style:         Spiral descent
#    Mapping sonar control enabled:
#                       Multibeam enabled
#                         Multibeam receive gain:         83
#                         Multibeam transmit gain:        220
#                         Multibeam minimum range fraction: 0.3
#                       Subbottom enabled
#                       Low sidescan enabled
#                       High sidescan disabled
#
# Mission Parameters:
#    Vehicle Speed:         1.500000 (m/s) 2.915769 (knots)
#    Desired Vehicle Altitude: 75 (m)
```

```
#    Minimum Vehicle Altitude: 75 (m)
#    Abort Vehicle Altitude:   30 (m)
#    Maximum Vehicle Depth:    2525.307922 (m)
#    Abort Vehicle Depth:      2562.807922 (m)
#    Descent Vehicle Depth:    3 (m)
#    Spiral descent depth:     2324.917643 m
#    Spiral descent altitude:  50 m
#    Forward Looking Distance:  (m)
#    Waypoint Spacing:         100 (m)
#    GPS Duration:             600 (s)
#    Descend Rate:             0.417 (m/s)
#    Ascend Rate:              1 (m/s)
#    Initial descend Duration: 300 (s)
#    Setpoint Duration:        30 (s)
#
# The primary waypoints from the route file are:
#  <number> <longitude (deg)> <latitude (deg)> <topography (m)> <distance (m)> <type>
#  0 −129.588618 46.504590 −2384.917643 0.000000 3
#  1 −129.583151 46.507559 −2412.977865 533.709482 3
#  2 −129.569223 46.503420 −2548.389974 1697.143568 1
#  3 −129.566359 46.501080 −2494.963053 2037.557099 3
#  4 −129.548611 46.529852 −2539.510864 5512.537193 4
#  5 −129.551250 46.530628 −2562.807922 5732.537193 3
#  6 −129.568962 46.501924 −2531.034424 9199.497998 4
#  7 −129.571600 46.502699 −2519.138489 9419.497998 3
#  8 −129.553889 46.531404 −2547.114624 12886.458803 4
#  9 −129.556529 46.532180 −2514.533569 13106.458803 3
#  10 −129.574238 46.503475 −2470.815735 16573.419607 4
#  11 −129.576876 46.504250 −2444.596313 16793.419607 3
#  12 −129.559168 46.532956 −2521.781921 20260.380412 4
#  13 −129.561807 46.533732 −2537.382141 20480.380412 3
#  14 −129.579514 46.505026 −2429.459961 23947.341216 4
#  15 −129.582152 46.505801 −2412.764343 24167.341217 3
#  16 −129.564447 46.534508 −2545.397705 27634.302021 4
#  17 −129.567086 46.535284 −2534.068665 27854.302021 3
#  18 −129.584791 46.506576 −2398.283020 31321.262826 4
#  19 −129.587429 46.507352 −2390.671509 31541.262826 3
#  20 −129.569726 46.536059 −2489.889282 35008.223630 4
#  21 −129.572365 46.536835 −2465.280823 35228.223630 3
#  22 −129.590068 46.508127 −2389.067017 38695.184435 4
#  23 −129.592706 46.508902 −2409.290771 38915.184435 3
#  24 −129.575005 46.537610 −2466.533142 42382.145240 4
#  25 −129.577645 46.538386 −2491.371094 42602.145240 3
#  26 −129.595345 46.509677 −2397.609253 46069.106044 4
#  27 −129.597984 46.510452 −2413.315918 46289.106044 3
#  28 −129.580285 46.539161 −2499.048889 49756.066849 4
#  29 −129.582925 46.539937 −2523.030640 49976.066849 3
#  30 −129.600622 46.511227 −2443.481018 53443.027653 4
#  31 −129.603261 46.512002 −2419.008240 53663.027653 3
#  32 −129.585565 46.540712 −2518.522400 57129.988458 4
#  33 −129.588205 46.541487 −2498.521301 57349.988458 3
#  34 −129.605900 46.512777 −2450.386536 60816.949263 4
#  35 −129.608539 46.513552 −2473.623230 61036.949263 3
```

```
#   36 −129.590845 46.542262 −2487.422180 64503.910067 4
#   37 −129.593485 46.543038 −2491.040466 64723.910067 3
#   38 −129.611178 46.514327 −2472.610657 68190.870872 3
#   39 −129.606711 46.517999 −2485.114583 68723.704236 3
#   40 −129.557338 46.509809 −2488.398743 72619.152031 3
#   41 −129.550415 46.521262 −2545.285828 73998.189601 3
#   42 −129.600724 46.536447 −2470.920736 78209.372536 3
#   43 −129.606972 46.525648 −2443.988281 79501.503455 4
#
# A total of 550 mission points have been defined.
#
# Define Mission parameters:
#define MISSION_SPEED      1.500000
#define MISSION_DISTANCE   79501.503455
#define MISSION_TIME       57203
#define MISSION_TIMEOUT    60067
#define DEPTH_MAX          2525.307922
#define DEPTH_ABORT        2562.807922
#define ALTITUDE_DESIRED   75.000000
#define ALTITUDE_MIN       75.000000
#define ALTITUDE_ABORT     30.000000
#define GPS_DURATION       600
#define DESCENT_DEPTH      3.000000
#define SPIRAL_DESCENT_DEPTH      2324.917643
#define SPIRAL_DESCENT_ALTITUDE   50.000000
#define DESCEND_DURATION   300
#define SETPOINT_DURATION  30
#define GPSMINHITS         10
#define ASCENDRUDDER       3.000000
#define ASCENDPITCH        45.000000
#define ASCENDENDDEPTH     2.000000
#define DESCENDRUDDER      3.000000
#define DESCENDPITCH       −30.000000
#define MAXCROSSTRACKERROR 30
#define RESON_DURATION     6
#q
```

SEE ALSO

mbsystem(1), **mbgrdviz**(1)

BUGS

Perhaps.